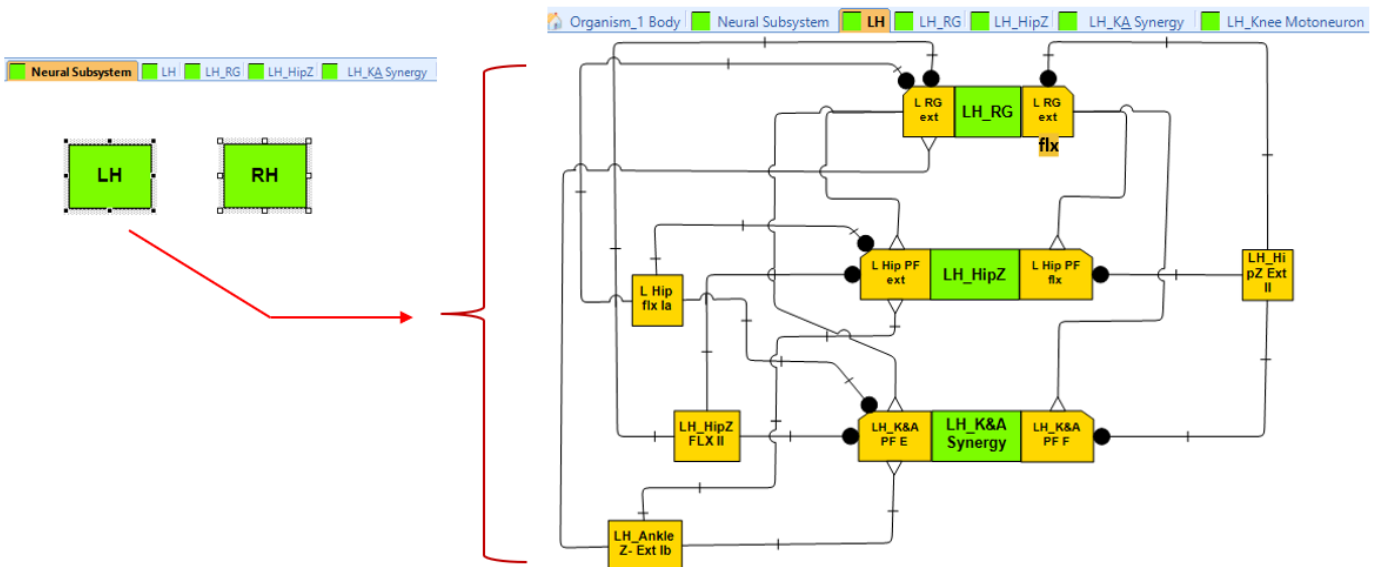
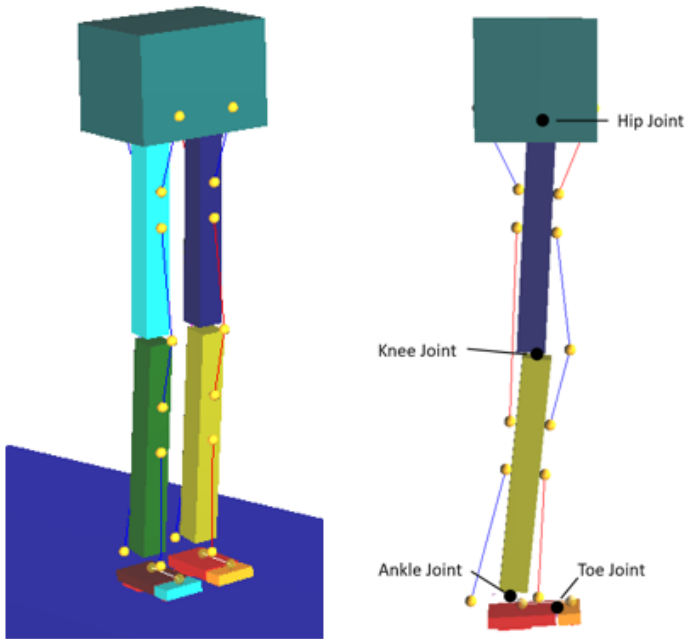


AnimatLab: body and controller hierarchy

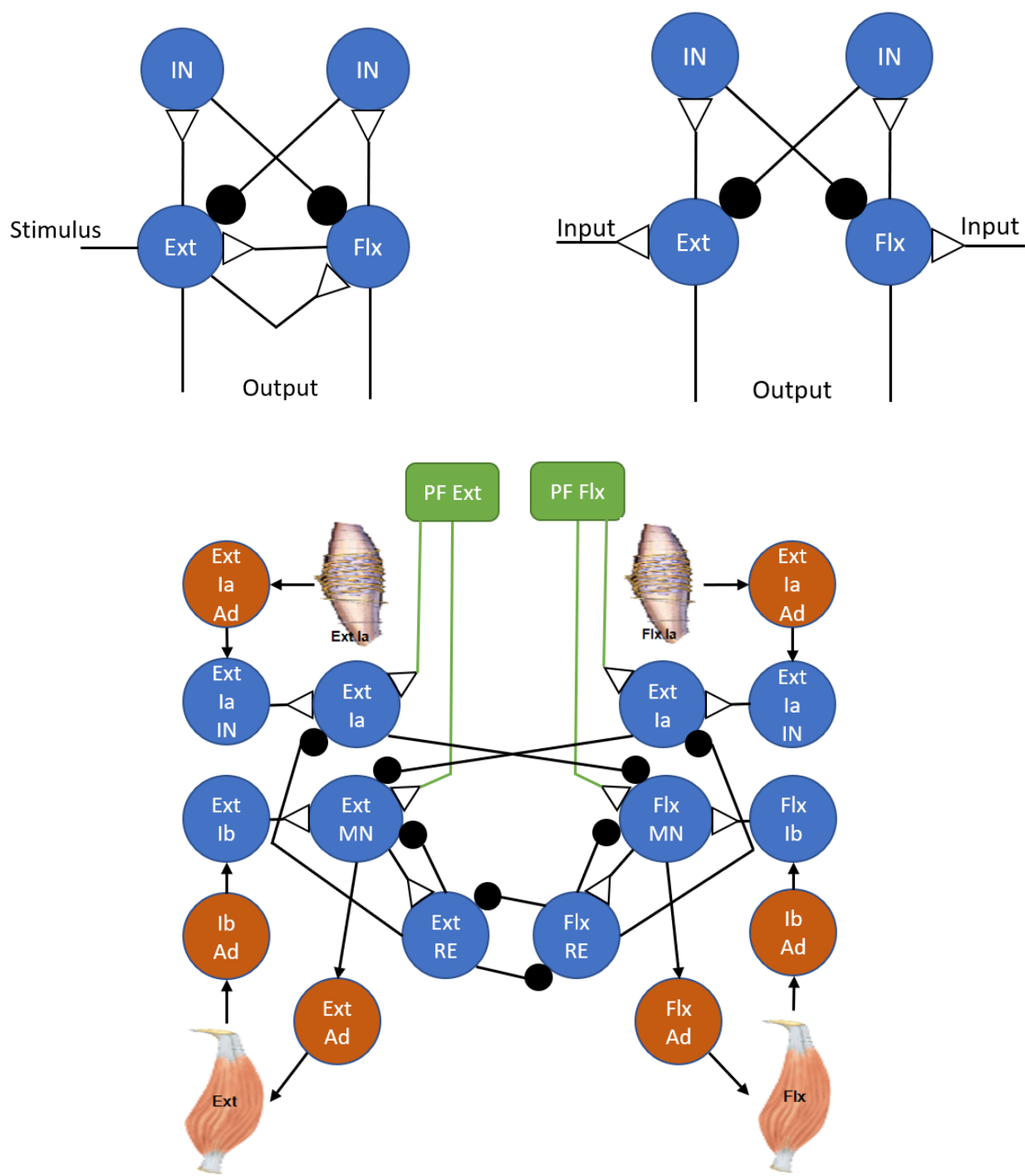
Figure review | dissertation placement: Methods unless identified as Results



The supplied body and subsystem illustrations show the segmented walker, its joints, and left/right neural hierarchy. These are architecture illustrations, not images from the new phase-1 trajectory.

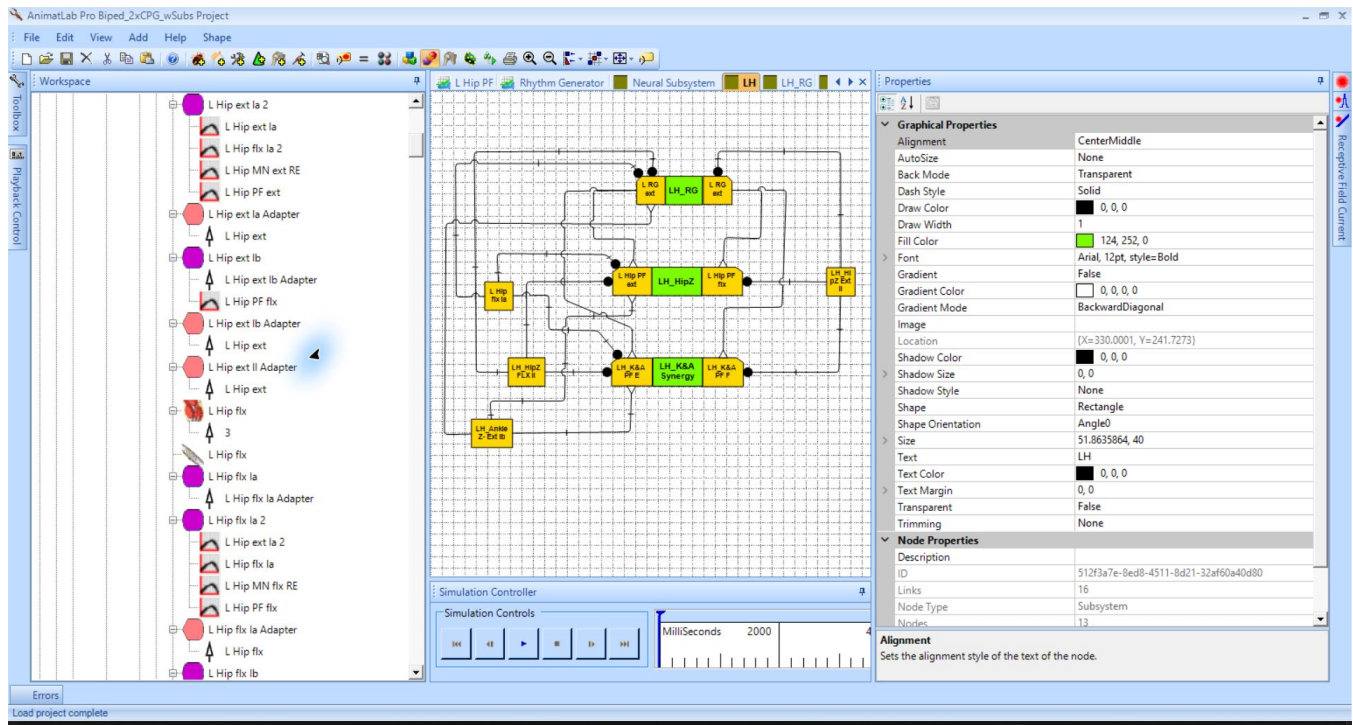
AnimatLab: neural circuit organization

Figure review | dissertation placement: Methods unless identified as Results



Reference schematics: rhythm generation (upper left), pattern formation (upper right), and sensorimotor pathways (below). The source labels and synaptic symbols are preserved. These show controller organization, not newly validated simulation results.

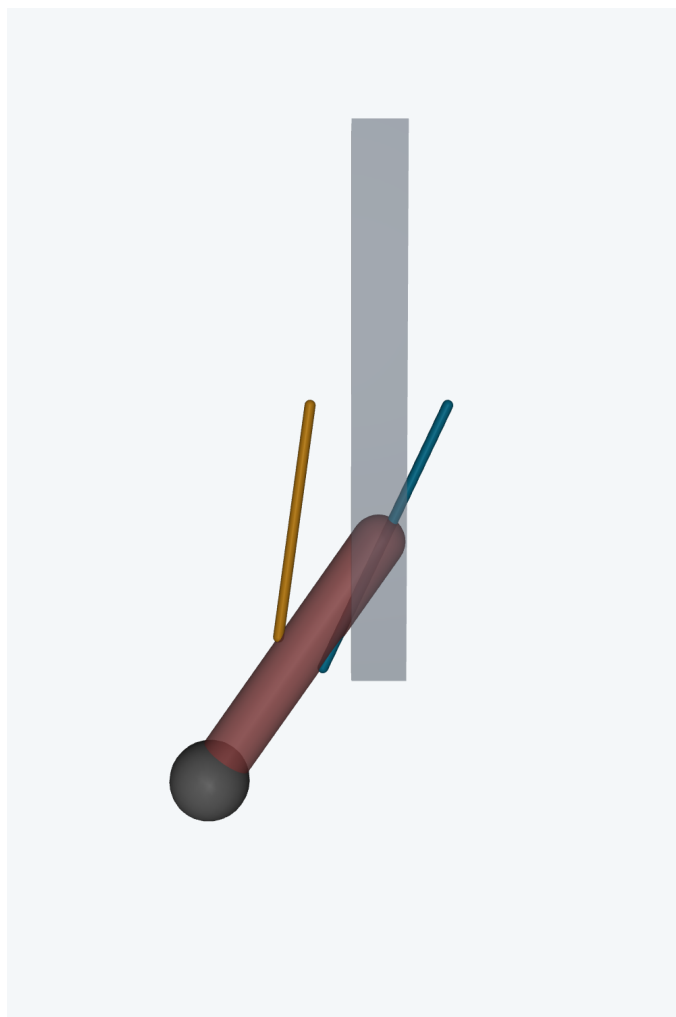
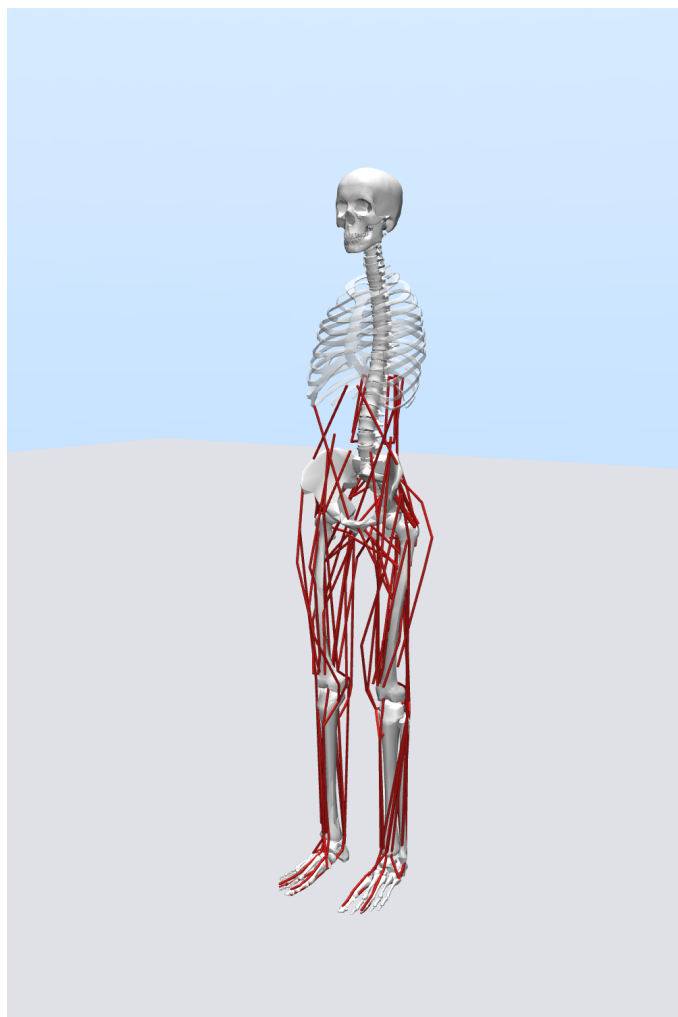
Figure review | dissertation placement: Methods unless identified as Results



Layout preview; final LaTeX pagination and compilation remain to be checked in Overleaf.

MuJoCo: converted robot and BPA knee

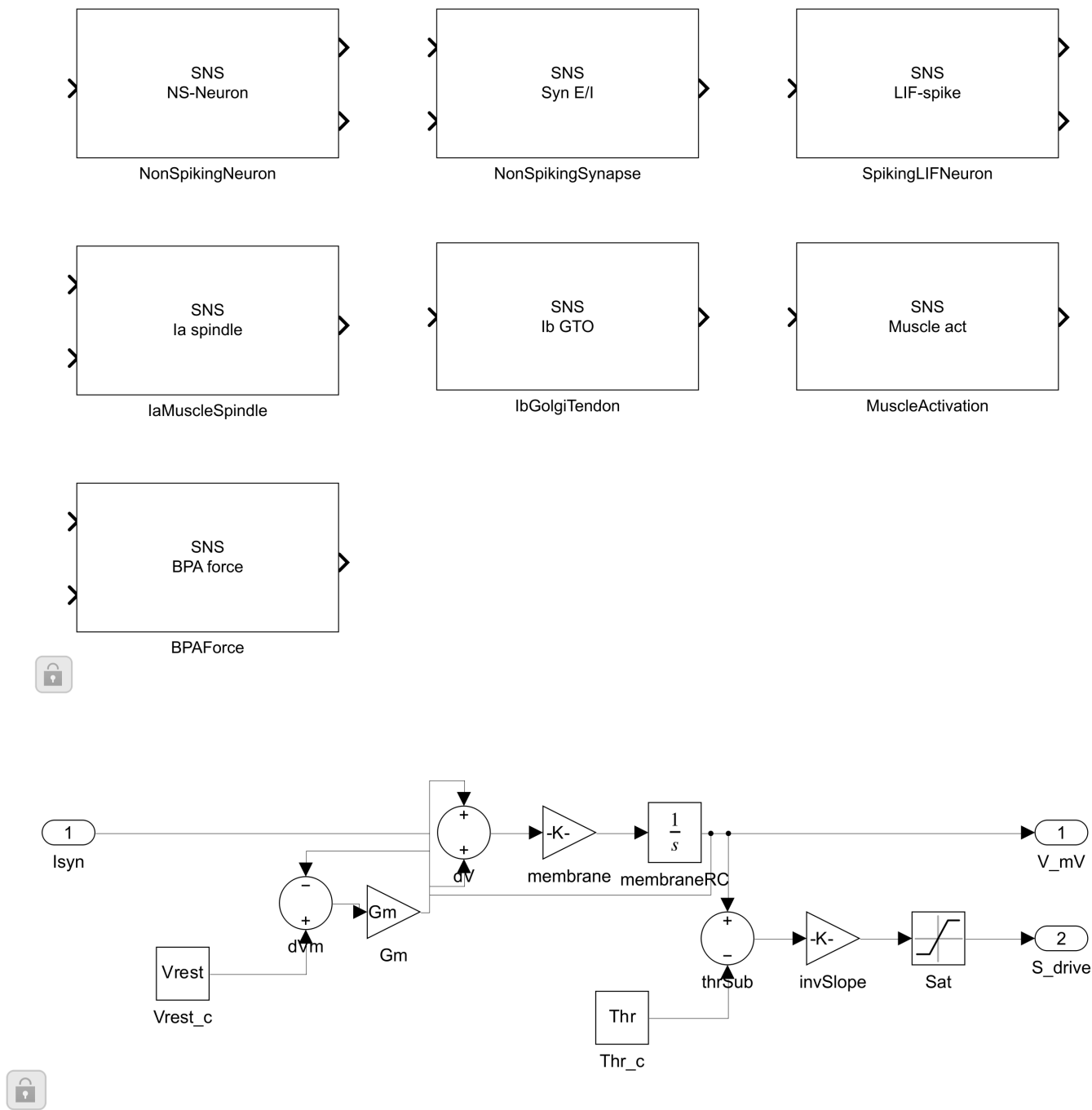
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Native MuJoCo renders. Left: converted Gait2392 robot at keyframe 0 with muscle paths. Right: the independent synthetic knee at an illustrative static -35-degree pose, showing cyan flexor and gold extensor routes. Different scales. No claim of SNS control of the converted robot or validated torque.

Simulink: SNS library and neuron

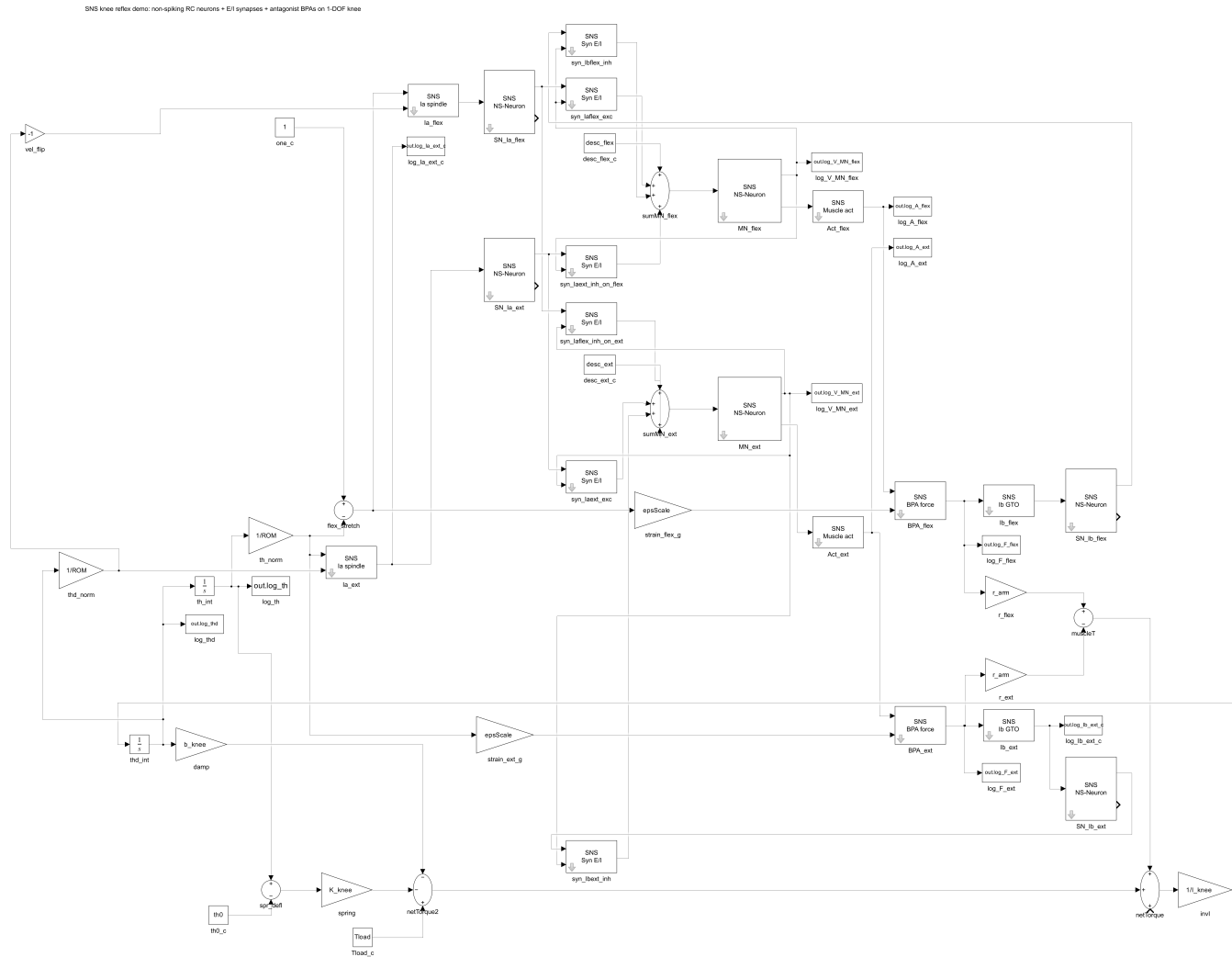
Figure review | dissertation placement: Methods unless identified as Results



Native high-resolution exports of all seven saved masked blocks and the non-spiking neuron implementation. The library is arranged for display. The BPA force block and plant parameters remain provisional; these diagrams do not establish quantitative regulation or a CAD-coupled Simscape Multibody plant.

Simulink: complete knee-reflex topology

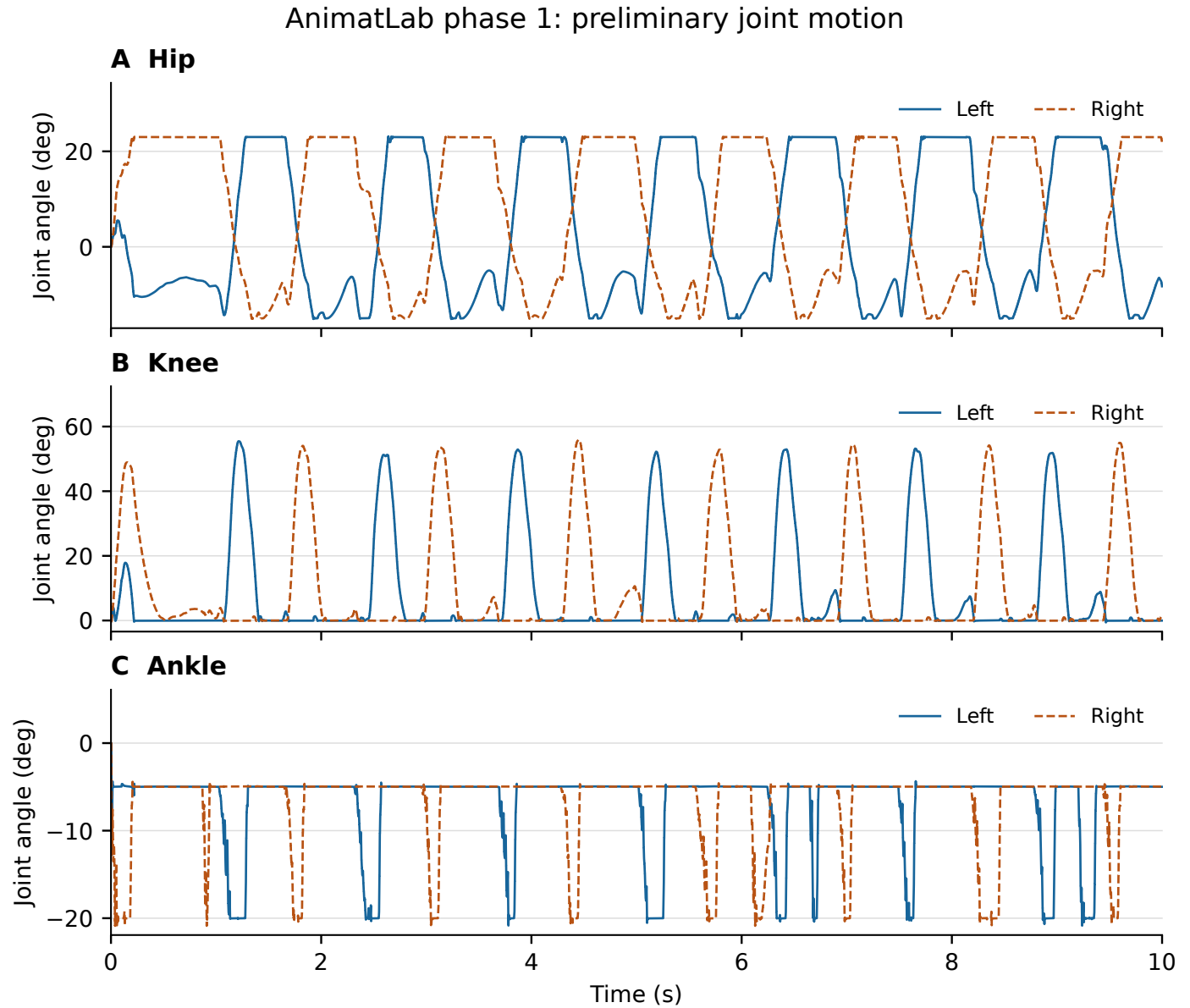
Figure review | dissertation placement: Methods unless identified as Results



Native high-resolution export of the complete reduced-order reflex model. The display was automatically arranged and every input port's source block and port were checked against the original. Source models were not saved. Mechanical and BPA force parameters remain provisional.

Preliminary Results: AnimatLab joint motion

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Bilateral hip, knee, and ankle angles from the new phase-1 run. All 50,001 samples from 0 through 10 s are retained without filtering; nine trailing zero-padding rows were excluded. The model applied an external horizontal force during the first second. Stable walking and effective ground-contact gating are not established.